

PRANAV V

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CAREER OBJECTIVE

Machine Learning enthusiast skilled in Python, SQL, and full-stack development using React. Experienced in building data science pipelines and implementing machine learning algorithms. Passionate about leveraging AI to solve real-world challenges, with a strong foundation in deep learning, generative models, and applied AI research.

EDUCATIONAL QUALIFICATION

- **Bachelor of Engineering – Artificial Intelligence and Machine Learning**
CMR Institute of Technology, AECS Layout, Bengaluru
CGPA: 8.66, 2026
- **Pre-University Course - Science (PCMB)**
St Joseph's Pre University College, Museum Road, Bengaluru
83.3%, 2022
- **10th Grade – ICSE**
Cambridge School, K R Puram, Bengaluru
81.8%, 2020

TECHNICAL SKILLS

- **Programming Languages:** Python, Java (Intermediate)
- **Frameworks:** React.js
- **Software & Analytics:** Visual Studio Code, Jupyter Notebook, PowerBI, Tableau, Android Studio, MS Excel
- **Operating Systems:** Windows
- **Web Technology:** HTML
- **Machine Learning and AI:** OpenCV, NLP, TensorFlow
- **Databases:** SQL (Intermediate)
- **Cloud Platforms:** AWS (Basic)
- **Gen-AI Tools:** OpenAI, GenAI, LLM's, RAG
- **Version Control:** Git and GitHub

PROJECTS

CardioGuard: AI-Powered Mobile Heart Disease Analyzer

- **Description:** Heart Sound Analyzer is an AI-based system that transforms any smartphone into a clinical-grade cardiac screening tool. The project evolved through three phases: core AI development, web deployment, and mobile optimization. Starting with the PhysioNet 2016 dataset (3,240+ heart sound recordings), the system applies advanced audio preprocessing (mel-spectrograms) and deep learning to achieve 92.23% AUC accuracy. Optimized with TensorFlow Lite, the model size was reduced by 91% (1.33MB → 0.12MB) and inference speed improved, enabling real-time analysis directly on smartphones. The web app (Streamlit) supports multi-device access, QR-based workflows, and cross-platform audio input. Designed for accessibility, it provides offline, mobile-first cardiac screening with clinical-grade reliability
- **Tech Stack:** Python, TensorFlow/Keras, TensorFlow Lite, Librosa, NumPy, Pandas, Scikit-learn, Streamlit, OpenCV, Matplotlib, Seaborn

Operation Gridlock: GEN-AI-Powered Vehicle Detection & Pursuit System (Gen-AI)

- **Description:** Operation Gridlock is a real-time AI system that detects, identifies, and tracks vehicles across citywide CCTV networks. Using segmentation-level precision (SAM3), high-detail visual extraction (Qwen ImageEdit 2509), and clarity restoration (Real-ESRGAN + Stable Diffusion), the system scans heavy traffic, reconstructs unclear visuals, and isolates the target vehicle with high accuracy. A route-forecasting engine predicts the thief's

next junction and ETA, triggering selective camera activation and generating instant geo-coordinate alerts. Designed for law enforcement automation, Operation Gridlock reduces surveillance overload and enables intelligent, fast vehicle interception.

- **Tech Stack:** SAM3(Meta), Qwen ImageEdit 2509 (7B), LIAMA 3.2, Real-ESRGAN, Stable Diffusion, CCTV feed processing, real-time alerting dashboard.

AI Resume Screening System with Chatbot Integration

- **Description:** Developing a comprehensive web application to revolutionize recruitment by automating resume screening and enhancing candidate engagement. This system leverages advanced AI/NLP to accurately parse resumes, semantically match candidates to job descriptions, and provide explainable, bias-mitigated scores. It features a proactive, AI-driven chatbot offering real-time resume feedback to candidates before submission, alongside robust HR tools for collaborative evaluation and intelligent insights.
- **Tech Stack:** React.js, Node.js (Express), AI/NLP (Hugging Face, spaCy), Supabase, Auth.js, Rasa Open Source.

INTERNSHIPS

- **Company Name:** Cheerans Global Solutions Pvt Ltd.
- **Duration:** 4 Months (January 2026 – May 2026)
- **Designation:** Data Scientist Intern
- **Responsibilities:**
 - Executed tasks in data analysis, data cleaning, and created interactive dashboards
 - Designed and implemented the Frontend for a Billing Software for a café.

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

Hackathons/Competitions

- Participated and secured 2nd place in the “National Level Deep Learning Hackathon” organized by BGSIT, 2025.
- Participated in the “Gen AI Hackathon” organized by Trendy Tech, 2025.
- Participated in the “Ops-Fusion Hackathon” organized by CMRIT, 2025.
- Participated in the “Vyuhattech 2.0” organized by CMRIT, 2025.

Seminars and Workshops

- Attended a workshop on “Patent Drafting” organized by the Department of Artificial Intelligence and Machine Learning in association with CMRIT Institution’s Innovation Council.
- Attended a seminar on “Generative AI” organized by CMR Institute of Technology.
- Attended a seminar on “Startup and Innovation” organized by CMR Institute of Technology.

Patents

- AI Powered Pet Monitoring System - Developed a smart system using AI and computer vision to monitor pet behaviour, detect health anomalies, and send real-time alerts to owners. Filed for patent protection to secure innovation in intelligent pet care technology.

NSS

- Actively participated in orphanage visits, engaging with children through interactive activities, distributing books, stationery, and meals, and organizing sessions that encouraged learning and emotional support.

Technical Clubs

- Member of Cybernauts, club of the Department of Artificial Intelligence and Machine Learning.

AWARDS AND ACHIEVEMENTS

- Received a Certificate of Appreciation for designing posters for the “Open Day” event held on 17th May 2025 organized by CMR Institute of Technology.
- Served as the Class Representative in the 1st and 2nd semesters.

PERSONAL DETAILS

Date of Birth	: 12/10/2004
Gender	: Male
Nationality	: Indian
Linguistic Competencies	: English, Kannada (Native), Hindi, Telugu
Hobbies	: Travelling, Swimming, Badminton, Bike Riding, Cooking